Arista Access Node Failure SOP (KOR-ACC-79) - Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss)

Node: KOR-ACC-79 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_7000

Description

The Access Node (KOR-ACC-79) failure within the Koramangala Datacenter, utilizing Arista equipment, is typically attributed to Hardware/Environmental Faults primarily affecting the ASIC memory parity errors, line card crashes, or power supply redundancy loss. This failure domain is more prevalent in roles with high traffic volumes or those with an increased reliance on hardware redundancy to ensure continuous operation. The underlying protocol mechanics governing this failure domain are rooted in the Arista's Extensible Operating System (EOS), which relies heavily on the ASIC (Application-Specific Integrated Circuit) hardware to process and forward data packets. In scenarios where the ASIC memory parity errors occur, the EOS may experience difficulties in maintaining packet consistency, leading to potential crashes or line card failures. Furthermore, power supply redundancy loss can also impact the hardware's ability to function, resulting in the node becoming non-operational. Hence, this SOP is designed to systematically diagnose and rectify the hardware-related issues affecting the Arista Access Node (KOR-ACC-79) within the Koramangala Datacenter.

Pre-requisite Checks

- Pre-requisite 1
- Pre-requisite 2
- Pre-requisite 3

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: System resource utilization (e.g., CPU, memory, and buffer usage), Description: This command provides an overview of the system's resource utilization, which can help identify potential issues related to ASIC memory parity errors or line card crashes.
```

```
Command: show interface, Expected Output: Interface information (e.g., speed, duplex, and link status), Description: This command displays interface information, which can help diagnose issues related to line card crashes or power supply redundancy loss.

Command: show power supply, Expected Output: Power supply status and redundancy information, Description: This command provides power supply status and redundancy information, which can help diagnose issues related to power supply redundancy loss.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card, Command: replace asic slot 1, Description: This step involves replacing the faulty ASIC or line card with a new one, which can be obtained from the Arista's spare parts inventory. Ensure that the replacement is done in a controlled environment to prevent any further damage to the equipment.

Step 2: Step: Reboot the node, Command: reboot, Description: This step involves rebooting the node to ensure that the new ASIC or line card is properly initialized and configured. Ensure that the node is properly powered and the redundant links are functioning before rebooting.

Step 3: Step: Verify the node's operation, Command: show system resource, Description: This step involves verifying that the node is operating properly and that the issues related to ASIC memory parity errors, line card crashes, or power supply redundancy loss have been resolved.

Post-Check Verification

```
Command: show system resource, Expected Output: System resource utilization (e.g., CPU, memory, and buffer usage) within normal limits, Description: This command provides an overview of the system's resource utilization, which can help verify that the issues related to ASIC memory parity errors or line card crashes have been resolved.

Command: show interface, Expected Output: Interface information (e.g., speed, duplex, and link status) within normal limits, Description: This command displays interface information, which can help verify that the issues related to line card crashes or power supply redundancy loss have been resolved.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy startup-config running-config, Description: This step involves restoring the original configuration to the node, which can help prevent any further damage to the equipment.

- Step: Reboot the node, Command: reboot, Description: This step involves rebooting the node to ensure that the original configuration is properly initialized and configured.